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EEE 414 — Past Questions & Materials

by Renance

Course: EEE 414

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 2

Levels: 400

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Paper 1 of 2 — EEE 414 PQ solutions-1

Level 400 • First Semester • PastQuestion

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EEE 414.2019

V₁ - The voltage applied to the primary of a single phase xformer at no-load is given by $v = 200 \cos \omega t + 80 \cos 3\omega t$, if the primary has 300 turns, then calc. the maximum value of the flux in the core for a fundamental freq of 50 Hz.

2 - A D.C. compound motor having a rating of 10 kW, 1150 rpm, 230 V, 50 A, has the following losses at full load:
 Bearing friction loss = 40 W; Brush friction loss = 50 W; windage loss = 200 W

i. Total mech. losses = 290 W

ii. Iron losses = 420 W

iii. Copper loss in the shunt field = 120 W

iv. Copper loss in

a Armature = 500 W (b) Series field = 75 W (c) Commutator ring winding = 70 W

v. Total Cu loss in the armature set = 595 W

Calc. the losses & efficiency at no load & at 80% of the normal rating of the D.C. compound motor.

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✓ 3 - A 4-pole wave wound motor armature has 880 conductors & delivers 120 A. The brushes have been displaced through 3 angular degrees from the geometrical axis. Calc.

i. Demagnetizing amp-turns/pole

ii. Cross-magnetizing "

iii. The additional field current for neutralizing the demagnetization of the field winding having 1100 turns/pole.

$\frac{250}{500}$
 $20 \left(\frac{1}{2} \right)$

✓ 4 - The following test data were obtained from a 1/4 hp, 1 ϕ , 120V, 60Hz, 1730 rpm induction motor, with stator winding resistance $\approx 2.9 \Omega$.

No. of poles = 4, $f_{sc} = 4$

Blocked rotor test:

$V = 43V$, $I = 5A$, $P = 140W$

No-load test:

$V = 120V$, $I = 3.5A$, $P = 125W$.

Determine the i/p current, i/p power, power factor, developed torque, i/p power, efficiency of the motor, air-gap power

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10 marks

and rotor copper loss of the motor is running at the rated speed when connected to a 120V supply, where the magnetizing reactance is 55.72Ω , leakage reactance of the rotor referred to the stator is 3.26Ω , resistance of rotor referred to the stator is 2.7Ω .

5- Show mathematically that the MMF along a position θ is equal to the forward & backward rotating MMFs.

6- Two alternators running in parallel supply lighting load of 2500kW & a motor load of 5000kW at 0.707 PF. one machine is ~~not~~ loaded to 4000kW at a P.F. of 0.8 lagging, what is the kW of & P.F. of the other machine?

7- Two transformers operating in parallel have the following characteristics

No.1 - % $I_R = 1.0\%$ & % $I_X = 5.0\%$

No.2 - % $I_R = 1.5\%$ & % $I_X = 4.0\%$

How will they share a load of 100kVA at 0.8 p.f. lagging?

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8 - Determine the main dimensions of the core for a 250kVA, 11/0.415kV, 50 Hz, three-phase xformer with star connected windings. Assume the following data: Voltage per turn, 9; maximum flux density, 1.25 T ; $A_c = 0.62 \text{ m}^2$; window space factor $k_w = 0.27$; window height/window width = 2. Current density, $250 \times 10^4 \text{ A/m}^2$ & Stack factor of 0.9.

9 - Determine the core area, no of turns & the position of the tapping point of a 500kVA, 50 Hz single phase 6600/5000 V auto transformer, assuming the following approximate values: emf per turn 8V, maximum flux density 1.3 Wb/m^2 .

10 - Determine the no of turns on each commutating pole of a 6-pole machine, if the flux density in the air-gap of the commutating pole = 0.5 Wb/m^2 at full load & the effective length of the air-gap

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is 4mm. The full load current is 500 A & the armature is lap-wound with 540 conductors. Assume the ampere turns required for the remainder of the magnetic circuit to be one-tenth of the air-gap.

- 11— Show that the power developed in a salient-pole synchronous generator is given by

$$P_d = \frac{V^2(X_d - X_q)}{2X_d X_q} \sin 2\delta \text{ per phase,}$$

- 12— Show that the mech. power developed by a synchronous motor is given by

$$P_{\text{max}} = \frac{VE_b}{X_s} \text{ per phase}$$

- 3— ~~Show mathematically that the max. along a position δ is equal to the forward & backward rotating m.f.s.~~

5, 8, 11, 12

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①

Solution is

1. $V = 200 \cos \omega t + 80 \cos 3\omega t$

No. of turns $N_1 = 300$

Frequency $f = 50 \text{ Hz}$

Max value of flux in the core $\phi_{\text{max}} = ?$

Recall,

$\phi_{\text{max}} = \text{Peak value of the fundamental component of the flux} - \text{Peak value of the third harmonic component of the flux.}$

i.e. $\phi_{\text{max}} = \phi_{1\text{max}} - \phi_{3\text{max}}$

But

$$\phi_{1\text{max}} = \frac{E_1}{4.44 f N_1} \quad \text{and} \quad \phi_{3\text{max}} = \frac{E_3}{3 \times 4.44 f N_1}$$

where,

$$E_1 = \text{RMS value of fundamental component of emf} = \frac{\text{Amplitude}}{\sqrt{2}}$$

$$E_3 = \text{RMS value of third harmonic component of emf} = \frac{\text{Amplitude}}{3\sqrt{2}}$$

$$\therefore E_1 = \frac{200}{\sqrt{2}} = 141.42 \text{ V} \quad \text{and} \quad E_3 = \frac{80}{3\sqrt{2}} = 58.57 \text{ V}$$

And also,

$$\phi_{1\text{max}} = \frac{E_1}{4.44 f N_1} = \frac{141.42}{4.44 \times 50 \times 300} = 2.1234 \text{ mwb}$$

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(2)

$$\phi_{3max} = \frac{E_3}{3 \times 4.44 f N_3} = \frac{56.57}{3 \times 4.44 \times 50 \times 300} = 0.2831 \text{ mwb}$$

Therefore,

$$\begin{aligned} \phi_{max} &= \phi_{1max} - \phi_{3max} \\ &= (2.1234 - 0.2831) \text{ m} \\ &= \underline{1.8401 \text{ mwb}} \end{aligned}$$

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(3)

THREENo of poles $(p) = 4$ polesArmature Conductors $Z = 880$ conductors.Angle displaced $\theta_m = 3^\circ$.Current $I_a = 120$ A.

$$\text{But } I = \frac{I_a}{p} = \frac{120}{2} = 60 \text{ A.}$$

i. Demagnetizing amp-turns/pole $(A_{td}/p) = \frac{ZI \theta_m}{360}$

$$= \frac{880 \times 60 \times 3}{360}$$

$$= 440 \text{ AT}$$

ii. Cross-magnetizing amp-turns/pole $(A_{tc}/p) = \frac{ZI \left[\frac{1}{2p} - \frac{\theta_m}{360} \right]}$

$$= 880 \times 60 \left[\frac{1}{2 \times 4} - \frac{3}{360} \right]$$

$$= 52800 \left[0.125 - 0.00833 \right]$$

$$= 52800 \left[0.1167 \right]$$

$$= 6160 \text{ AT}$$

iii. Additional field current = Armature Demagnetization
 Demagnetization of the field winding

$$= \frac{440}{1100}$$

$$= \underline{\underline{0.4A}}$$

SEVEN

$$Z_A = (1 + j5)\Omega = 5.1 \angle 78.7^\circ$$

$$Z_B = (1.5 + j4)\Omega = 4.3 \angle 69.4^\circ$$

Load in kVA $S = 100 \text{ kVA}$

Power factor $\cos \phi = 0.8$

Hence, total load to be shared = $100 \angle -36.87^\circ$
 (lagging)

$$\therefore S_A = S \cdot \frac{Z_B}{Z_A + Z_B} = \frac{100 \angle -36.87^\circ \times 4.3 \angle 69.4^\circ}{5.1 \angle 78.7^\circ + 4.3 \angle 69.4^\circ}$$

$$S_A =$$

$$S_A = S \cdot \frac{Z_B}{Z_A + Z_B}$$

$$S_B = S \cdot \frac{Z_A}{Z_A + Z_B}$$

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$$Z_A + Z_B = (1 + j5) + (1.5 + j4) = 2.5 + j9 = 9.34 \angle 74.5^\circ$$

Therefore,

$$\begin{aligned} S_A &= S \cdot \frac{Z_B}{Z_A + Z_B} = \frac{100 \angle -36.87^\circ \times 4.3 \angle 69.4^\circ}{9.34 \angle 74.5^\circ} \\ &= \frac{430 \angle 32.53^\circ}{9.34 \angle 74.5^\circ} \\ &= 46.04 \angle -41.97^\circ = P \cos \phi = \underline{\underline{34.23 \text{ kW}}} \end{aligned}$$

$$\begin{aligned} S_B &= S \cdot \frac{Z_A}{Z_A + Z_B} = \frac{100 \angle -36.87^\circ \times 5.1 \angle 78.7^\circ}{9.34 \angle 74.5^\circ} \\ &= \frac{510 \angle 41.83^\circ}{9.34 \angle 74.5^\circ} \\ &= 54.60 \angle -32.67^\circ \\ &= P \cos \phi = \underline{\underline{45.96 \text{ kW}}} \end{aligned}$$

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(6)

SixFor 1st MachineLoad power (or kw) of lighting load, $P = 2500 \text{ kW}$

Load reactive power (or kVAR) of lighting load

$$Q = \frac{P \sin \phi}{\cos \phi}$$

$$\therefore \cos \phi = 1 \text{ for lighting load}$$

$$\therefore \phi = 0 \text{ and } \sin \phi = \sin 0 = 0$$

$$\therefore Q \text{ of lighting load} = \frac{2500 \times 0}{1} = 0$$

For 2nd Machine

P of motor load = 5000 kW

$$P.F. \cos \phi = 0.707$$

$$\therefore \sin \phi = 0.707$$

$$Q \text{ of motor load} = \frac{P \times \sin \phi}{\cos \phi} = \frac{5000 \times 0.707}{0.707}$$

$$= 5000 \text{ kVAR}$$

$$\text{Total load} = 2500 + 5000 = 7500 \text{ kW}$$

$$\text{Total kVAR} = 0 + 5000 = 5000 \text{ kVAR}$$

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(7)

Load sharing

Load taking by 1st machine = 4000 kW

$$Q \text{ of 1st machine} = \frac{4000 \times 0.6}{0.8} = 3000 \text{ kVAR}$$

2nd Machine will supply
 $= 7500 - 4000 = 3500 \text{ kW}$

$$Q = 5000 - 3000 = 2000 \text{ kVAR}$$

Since $Q = \frac{P}{\cos \phi} \times \sin \phi = P \tan \phi$

$$\therefore \tan \phi = \frac{Q}{P}$$

$$\phi = \tan^{-1} \left(\frac{Q}{P} \right) = \tan^{-1} \left(\frac{2000}{3500} \right)$$

$$\phi = \tan^{-1} 0.57143$$

$$\phi = 29.7448^\circ$$

$$\therefore \text{p.f., } \cos \phi = \cos 29.7448^\circ$$

$$= 0.8682 \text{ (aggr)} \\ \underline{\underline{}}$$

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⑧

$$N_{cp} I_a = \frac{Z I_c}{2P} + B_g l_g / \mu_0$$

where, N_{cp} = No of turns on the compole

I_a = Armature current

Z = No of armature conductors

I_c = Coil current

P = No of poles

l_g = Air-gap length under the compole

$Z = 540$, $I_a = 500 A$, $I_c = \frac{500 A}{6}$, $P = 6$

$$\therefore \text{Armature mmf} = \frac{Z \times I_a}{2} = \frac{540 \times \left(\frac{500}{6}\right)}{2} = 3750$$

$$\text{Compole air-gap mmf} = B_g \times l_g / \mu_0 = 0.5 \times 4 \times 10^{-3} / 4\pi \times 10^{-7} = 1591$$

mmf required for the rest of the magnetic ckt = 10% of 1591 = 159

$$\therefore \text{Total compole air-gap mmf} = 1591 + 159 = 1750$$

$$\text{Total mmf reqd} = 3750 + 1750 = 5500$$

$$\therefore N_{cp} I_a = 5500 \text{ or } N_{cp} = \frac{5500}{500} = 11$$

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(9)

$$N_{eq} = 2I_p / \mu + B_s L / \mu$$

NINE

$$E = 4.44 f \Phi_m N \text{ volt}$$

$$\Phi_m = \frac{E/N}{4.44f} = \frac{8}{4.44 \times 50} = 0.03604 \text{ Wb}$$

$$\text{Core area} = \frac{\Phi_m}{\text{Maximum flux density}} = \frac{0.03604}{1.3} = 0.0277 \text{ m}^2 = 2770 \text{ cm}^2$$

$$\text{Turns of H.V. side} = \frac{6600}{8} = 825$$

$$\text{Turns of LV side} = \frac{5000}{8} = 625$$

Hence, tapping should be 200 turns from high voltage end or 625 turns from the common end

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Two

No load losses = $420 + 290 + 140 = 830 \text{ W}$

The efficiency at no load is zero since no useful power is developed by the motor

At 80% load

At 80% of the nominal armature current, is approximately $\frac{80}{100} \left(\frac{4}{5} \right)$ of the full load

Losses in armature $\hat{=}$ $\left(\frac{80}{100} \right)^2 \times 595 = 380.8 \text{ W}$

No-load losses = 830 W

$\therefore$ Total losses = $380.8 + 830 = 1210.8 \text{ W}$

Useful power developed by the motor at 80%

$$P_o = \frac{80}{100} \times 10 \text{ kW} = 8 \text{ kW}$$

- Power supplied to the motor

$$P_i = P_o + \text{Total losses}$$

$$= 8000 + 1210.8$$

$$= 9210.8 \text{ W}$$

- Efficiency = $\frac{P_o}{P_i} \times 100$

$$= \frac{8000}{9210.8} \times 100 = \underline{\underline{86.85\%}}$$

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Given Data:

$$V = 120V \quad f = 60Hz \quad n_m = 1730rpm \quad p = 4$$

$$R_1 = 2.9\Omega$$

$$n_s = \frac{120f}{p} = \frac{120 \times 60}{4} = 1800rpm$$

$$s = \frac{n_s - n_m}{n_s} = \frac{1800 - 1730}{1800} = 0.039$$

For the Blocked Rotor Test

$$V_{bl} = 43V \quad I_{bl} = 5A \quad P_{bl} = 140W$$

$$Z_{bl} = \frac{V_{bl}}{I_{bl}} = \frac{43}{5} = 8.6\Omega$$

$$R_1 + R_2 = \frac{P_{bl}}{I_{bl}^2}$$

$$R_1 + R_2 = \frac{140}{5^2}$$

$$R_1 + R_2 = \frac{140}{25}$$

$$R_1 + R_2 = 5.6\Omega$$

$$\therefore R_2 = 5.6 - 2.9$$

$$R_2 = 2.7\Omega$$

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$$X_{bl} = \sqrt{(Z_{bl})^2 - (R_{l1} + R_{l2})^2}$$

$$= \sqrt{(8.6)^2 - (2.9 + 2.7)^2}$$

$$= \sqrt{(8.6)^2 - (5.6)^2}$$

$$= \sqrt{73.96 - 31.36}$$

$$= \sqrt{42.6}$$

$$X_{bl} = 6.527 \Omega$$

$$X_1 = X_2 = \frac{X_{bl}}{2} = \frac{6.527}{2} = 3.26 \Omega$$

For No-load Test

$$V_{nl} = 120V \quad I_{nl} = 3.5A \quad P_2 = 125W$$

$$Z_{nl} = \frac{V_{nl}}{I_{nl}} = \frac{120}{3.5} = 34.29 \Omega$$

$$X_{nl} = Z_{nl} \sin \phi_{nl}$$

$$\phi_{nl} = \cos^{-1} \left(\frac{P_{nl}}{V_{nl} I_{nl}} \right)$$

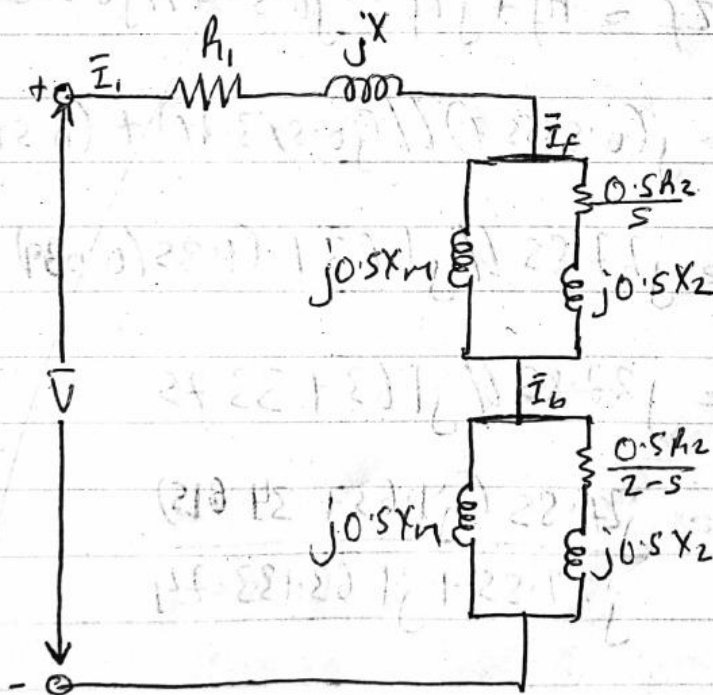
$$= \cos^{-1} \left(\frac{125}{120 \times 3.5} \right) = \cos^{-1}(0.2976) = 72.69^\circ$$

$$X_{nl} = Z_{nl} \sin \phi = 34.29 \sin(72.69) = 32.74 \Omega$$

$$X_m = 2 \left(X_0 - X_1 - \frac{X_2}{2} \right) = 2 \left(32.74 - 3.26 - \frac{3.26}{2} \right)$$

$$X_m = 2(27.85)$$

$$X_m = 55.7 \Omega$$



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$$\text{Input Current} = \frac{V_{in}}{(R_1 + R_4 + R_b) + j(X_1 + X_f + X_b)}$$

$$Z_f = R_f + jX_f = j0.5X_m // j0.5X_2 + 0.5R_2/s$$

$$Z_f = j(0.5 \times 85.7) // j(0.5 \times 3.26) + (0.5 \times 2.7) / 0.039$$

$$= j27.85 // j1.63 + (1.35 / 0.039)$$

$$= j27.85 // j1.63 + 34.615$$

$$= \frac{j27.85 (j1.63 + 34.615)}{j27.85 + j1.63 + 34.615}$$

$$Z_f = (12.987 + j16.789) \Omega$$

$$\therefore R_f = 12.987 \Omega \quad X_f = 16.789 \Omega$$

$$Z_b = R_b + jX_b = j0.5X_m // j0.5X_2 + 0.5R_2/(2-s)$$

$$Z_b = j(0.5 \times 85.7) // j(0.5 \times 3.26) + 0.5 \times 2.7 / (2 - 0.039)$$

$$= j27.85 // j1.63 + (1.35 / 1.961)$$

$$= j27.85 // j1.63 + 0.688$$

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$$Z_b = \frac{j27.85(j1.63 + 0.688)}{j27.85 + j1.63 + 0.688}$$

$$Z_b = (0.614 + j1.554)\Omega$$

$$R_b = 0.614\Omega \quad X_b = 1.554\Omega$$

$$\begin{aligned} \therefore I_{in} &= \frac{120}{(2.9 + 12.987 + 0.614) + j(3.26 + 16.789 + 1.554)} \\ &= \frac{120}{16.501 + j21.603} \end{aligned}$$

$$I_{in} = (2.6796 - j3.5081)A$$

$$I_{in} = (4.414 \angle -52.626^\circ)A$$

$$\begin{aligned} \text{— Input Power } P_{in} &= VI \cos \phi \\ &= 120 \times 4.414 \times \cos(-52.626^\circ) \\ &= 529.68 \times 0.6070 \\ &= \underline{\underline{321.52W}} \end{aligned}$$

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$$\text{Power factor} = \cos \phi = \cos(-54.626) \\ = \underline{\underline{0.607 \text{ (lagging)}}}$$

$$\text{Torque Developed } T = \frac{I^2 (R_f - R_b)}{\omega_s}$$

$$\text{But } \omega_s = \frac{2\pi n_s}{f} = \frac{2 \times 3.142 \times 1800}{60}$$

$$\omega_s = 188.52 \text{ rad/sec}$$

$$\therefore T = \frac{4.414^2 (12.987 - 0.614)}{188.52}$$

$$= \frac{19.483 (12.373)}{188.52}$$

$$= \underline{\underline{1.279 \text{ Nm}}}$$

$$\text{Output Power } P_o = P_{dev} - \text{Rotational losses} \\ P_{dev} = T \times \omega_s \times (1-s) = 1.279 \times 188.52 \times (1-0.039)$$

$$= 231.71 \text{ W}$$

$$\text{Rotational losses} = P_{cl} - I^2 n_l \left[R_1 + \frac{R_2}{4} \right]$$

$$= 125 - 3.5^2 \left[2.9 + \frac{2.7}{4} \right]$$

$$= 125 - 12.25 [3.575]$$

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$$= 125 - 43.794$$

$$\leq P_o = 231.71 - 81.206$$

$$= \underline{\underline{150.504 \text{ W}}}$$

- Efficiency of the motor $= \frac{P_o}{P_i} \times 100$

$$= \frac{150.504 \times 100}{321.52}$$

$$= \underline{\underline{46.81\%}}$$

- Air gap power $P_g = P_{gf} + P_{gl}$

$$= I^2 (R_f + R_b)$$

$$= 4.414^2 (12.987 + 0.614)$$

$$= 19.483 (13.601)$$

$$= \underline{\underline{264.988 \text{ W}}}$$

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~~Rotor Cu loss~~
 Air gap power rotor copper loss

$$P_{ACL} = sP_{sf} + (2-s)P_{sb}$$

$$= s[I^2 R_f] + (2-s)[I^2 R_b]$$

$$= (0.039 \times 4.414^2 \times 12.987) + (2 - 0.039) \times (4.414^2 \times 0.614)$$

$$= 9.868 + 23.459$$

$$= \underline{\underline{33.327 \text{ W}}}$$

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Paper 2 of 2 — EEE 414 PQ-1

Level 400 • First Semester • PastQuestion

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Synchronous Generators

1. State four(4) conditions necessary for parallel operation of a synchronous generator

Ans: 1. RMS line voltages of the two generators must be the same

2. Phase sequence must be the same

3. Phase angles of the corresponding phases must be the same

4. Frequency must be the same

2. What are the two(2) parameters that can be varied when a synchronous generator is connected to an infinite bus system?

Ans: $\Rightarrow$ the field excitation current, I_f
 $\Rightarrow$ the prime mover's mechanical torque, T

3. State the effects of each of the two(2) parameters mentioned in (2) above on the induced voltage, active and reactive power of the system

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E_o = induced voltage

Ans: synchronous generator

On the field excitation current, I_x

- impacts the induced voltage E_o
- does not affect the flow of active (real) power
- does cause reactive power to flow

On the ~~mechanical~~ prime mover's mechanical torque, T ,

- although both voltages have similar values, the phase angle produces a difference of potential across the synchronous reactance
- real (active) power will flow
- the rotor will accelerate, E_o will increase in value and begin to slip ahead of phasor E , leading by a phase angle δ

4. State four(4) steps to synchronize a generator

Ans:

- a) adjust the speed ~~set~~ regulator of the prime mover so that frequencies are close
- b) adjust the excitation so that generator voltage and system voltage are equal
- c) Observe the phase angle by means of a synchroscope, which indicates the phase angle between two voltages

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d) on the zero mark, the line circuit breaker is closed

5. State ^{four} ~~four~~ (4) conditions for synchronization of generators

Ans:

- the generator frequency is equal to the system frequency
- the generator voltage is equal to the system voltage
- the generator voltage is in phase with the system
- the phase sequence of the generator is the same as that of the system

6. State two (2) loads that synchronous generators depend on

Ans: Isolated loads and infinite bus

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Power triangle & Power factor

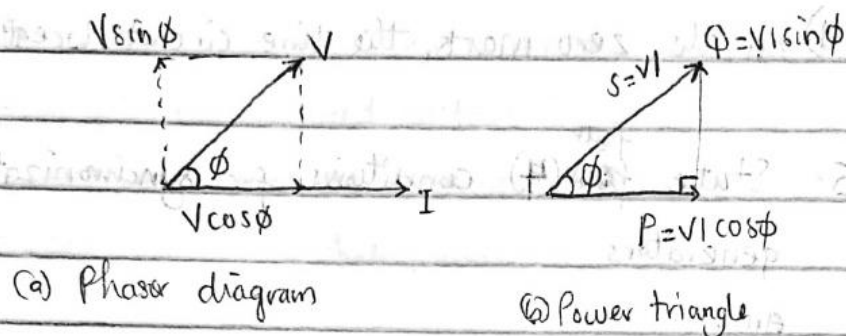


Figure (a) shows a phasor diagram in which the current I lags the applied voltage V by angle ϕ . The horizontal component of V is $V \cos \phi$ and the vertical component of V is $V \sin \phi$. If each of the voltage phasors is multiplied by the current I , figure (b) is obtained.

Apparent power, $S = VI$ voltamperes (VA)

True or active power, $P = VI \cos \phi$ watts (W)

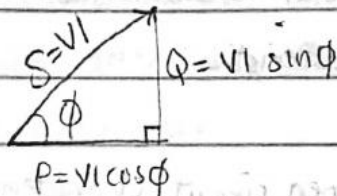
Reactive power; $Q = VI \sin \phi$ reactive voltamperes (Var)

$$\text{Power factor} = \frac{\text{True power, } P}{\text{Apparent power, } S}$$

$$\text{For sinusoidal voltages and currents, power factor} = \frac{P}{S} = \frac{VI \cos \phi}{VI}$$

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$$p.f = \cos \phi = \frac{R}{Z}$$



$$S = P + jQ$$

Apparent power active power reactive power

NB: A circuit in which current lags voltage (ie an inductive circuit) is said to have a lagging power factor, and indicates a lagging reactive power Q .

A circuit in which current leads voltage (ie capacitive circuit) is said to have a leading reactive power Q power factor, and indicates a leading reactive power Q .

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7. Briefly explain the following terms: -

- Magnetic or Open circuit Characteristic
- Internal or Total Characteristic
- External Characteristic

- Magnetic or Open circuit Characteristic - This characteristic, also known as no-load saturation characteristic, gives the relation between the generated emf in armature on no-load, E_o and the field current. If at a given speed. The shape of this curve is practically the same for all types of generators whether separately excited or self excited.

- Internal or Total Characteristic - This characteristic gives the relation between the emf actually generated in armature E (less than E_o due to the demagnetizing effect of armature reaction) and armature current. This curve, therefore, lies below the OCC.

- External Characteristic - This characteristic gives the relation between the terminal voltage V and the load current I under the given condition of speed and excitation.

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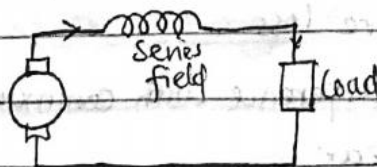
The terminal voltage V is always less than E on account of voltage drop in armature. This curve, therefore, lies below the internal characteristic. This characteristic is of utmost importance for determining the suitability of generators for a particular service.

8- Describe briefly the series-wound generator

The series-wound generator is connected in series with the armature winding so that the whole armature current flows through the field winding as well as the load, as shown in the figure below. According to figure below, it shows the connection

In the series wound generator, the field windings are joined in series with the armature conductors.

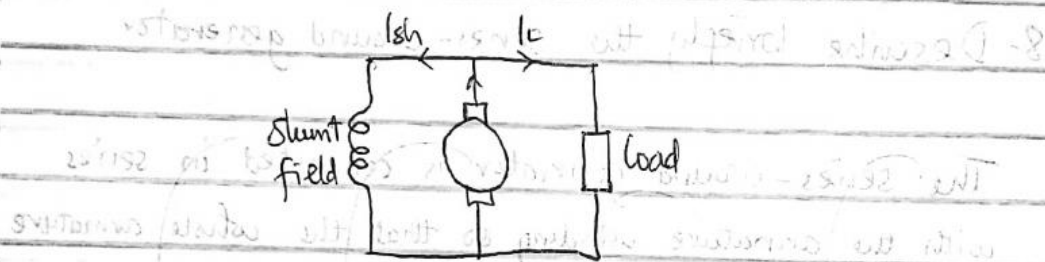
As they carry full load current, they consist of relatively few turns of thick wire or strips. Such generators are rarely used except for special purposes like as boosters etc.



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9. Describe briefly the shunt-wound generator

The field windings are connected across or in parallel with the armature conductors and have the full voltage of the generator applied across them.



The shunt field winding has many turns of fine wire having high resistance. Therefore only a part of armature current flows through the shunt field winding and the rest flows through the load.

10. State three (3) effects of harmonic currents in three-phase transformers.

- additional electrical (I^2R) loss owing to circulating currents
- increased core loss
- magnetic interference with communication circuits and protective gear

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11. State three effects of harmonic voltages in three phase transformers

- increased dielectric stresses
- electrostatic interference with communication circuits
- resonance between the inductance of the transformer windings and the capacitance of feeder to which these are connected.

12. State 2 ways in which armature reaction can be curtailed

- a) Compensating winding
- b) Interpoles or Compoles

13. Briefly describe the worst condition for connection of a transformer

The worst condition for connection of a transformer may occur when the applied voltage has zero value at the instant of switching, i.e. at $\alpha = -\pi/2$.

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14. State two (2) advantages of core-stepping in large size transformers.

1. It offers good mechanical strength
2. It is efficient for high frequencies
3. It has the advantage of preventing condensed flux leakage and iron loss.

15. State the three (3) conditions necessary for parallel operations of a single phase transformers

- a) Polarity
- b) Voltage Ratio
- c) Matched Impedances and Matched Impedance Angles

16. State the 5 conditions necessary for parallel operation of three phase transformers.

- a) Polarities of the transformer are the same
- b) The voltage ratings of both primary and secondary windings are identical
- c) Percent impedances of the transformers are the same in magnitude and have the same phase angle

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- d) Phase displacement between primary and secondary line voltages of the transformers are the same
- e) Phase sequence of the transformers are the same

17. What is saturable reactor? State the main use of a saturable reactor?

A saturable reactor is a special form of inductor where the magnetic core can be deliberately saturated by a direct electric current in a control winding.

The main use of a saturable reactor is to provide a very simple means to remotely and proportionally control the alternating current (AC) through a load such as an incandescent lamp; the AC is roughly proportional to the direct current (DC) in the control winding.

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18. State 3 necessities of parallel operation of transformers and the main disadvantage of parallel operation of transformers.

a) If the amount of power to be transformed is greater than that which can be handled by one transformer, it becomes necessary to employ two or more units in parallel.

b) To meet the additional demand on a power system it becomes necessary to either replace the existing unit by unit of greater capacity or to install additional unit to carry a portion of the load.

c) The cost of standby unit is reduced since spare transformer units are invariably required to ensure continuity of service in case of damage or some fault.

The main drawback is that available short-circuit current is increased, sometimes to a level beyond the interrupting capacity of the existing over-current devices.

A split-second bus with a tie breaker^{interlocked} is used to

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Prevent. parallel operation of transformer.

18. List five(5) methods of transformer cooling

- a) Air Natural (AN) cooling
- b) Air Blast. Cooling
- c) Oil Natural (ON) cooling
- d) Oil Natural Air Forced (OB) cooling
- e) Oil Natural Water Forced (OW) cooling

19. What are the general features of a.c armature winding?

20) State the equations for forward and backward field slip.

$$\text{Forward field slip, } S_f = \frac{n_s - n_m}{n_s} = S$$

$$\text{Backward field slip, } S_b = \frac{-n_s - n_m}{-n_s} = \frac{n_s + n_m}{n_s}$$

$$S_b = 2 - S$$

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21) What is slip?

Slip is the difference between the speed of the stator field (known as synchronous speed, N_s) and the actual speed of the rotor, N .

fraction slip, $S = \frac{N_s - N}{N_s}$

22) Explain briefly why the single-phase induction machine is not self-starting.

The single phase induction ~~not~~ machine is not self starting because if the rotor of such motor is given a push by hand or by another means in either direction, it will pick up speed and continue to rotate in the same direction developing operating torque.

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Calculations

- The voltage applied to the primary of a single phase transformer at no-load is given by $V = 200 \cos \omega t + 80 \cos 3\omega t$, If the primary has 300 turns, then calculate the maximum value of the flux in the core for a fundamental frequency of 50 Hz.

Solution

Given; $V = 200 \cos \omega t + 80 \cos 3\omega t$

$N_1 = 300$

$f = 50 \text{ Hz}$

$\Phi_{\text{max}} = ?$

RMS value of fundamental component of emf,

$$E_1 = \frac{200}{\sqrt{2}} = 100\sqrt{2} \text{ or } 141.42 \text{ V}$$

RMS value of third harmonic component of emf,

$$E_3 = \frac{80}{\sqrt{2}} = 40\sqrt{2} \text{ or } 56.57 \text{ V}$$

Peak value of fundamental component of flux,

$$\Phi_{1, \text{max}} = \frac{E_1}{4.44 f N_1} = \frac{141.42}{4.44 \times 50 \times 300} = 2.12 \text{ mWb}$$

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Peak value of third harmonic component of flux.

$$\Phi_{g \max} = \frac{E_3}{3 \times 4.44 f N_1} = \frac{56.57}{3 \times 4.44 \times 50 \times 300} = 0.283 \text{ mWb}$$

Maximum value of flux in the core,

$$\Phi_{\max} = \Phi_{1 \max} - \Phi_{g \max}$$

$$= 2.12 - 0.283$$

$$= 1.837 \text{ mWb}$$

2. A 4-pole wave wound motor armature has 880 conductors and delivers 120A. The brushes have been displaced through 3 angular degrees from the geometrical axis. Calculate

i) Demagnetizing amp-turns/pole

ii) Cross-magnetizing amp-turn/pole

iii) The additional field current for neutralizing the demagnetization of the field winding having 1100 turns/pole

Solution

Given, $Z = 880$

$$I = 120/2 = 60 \text{ A}$$

$$\theta = 3^\circ \text{ angular}$$

(i) Demagnetizing amp-turn/pole, $AT_d = 880 \times 60 \times \frac{3}{360}$

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$$AT_d = 440 \text{ AT} \quad \frac{ZI \frac{\theta_m}{360}}$$

ii) Cross-magnetizing amp-turns/pole, $ZI \left(\frac{1}{2p} - \frac{\theta_m}{360} \right)$

$$AT_c = 880 \times 60 \left(\frac{1}{8} - \frac{3}{360} \right) = 880 \times \frac{7}{60} \times 60$$

$$AT_c = 6,160$$

$$\text{Total AT/pole} = \frac{440 \times 60}{4} = 6600$$

$$\begin{aligned} \text{Hence, } AT_c/\text{pole} &= \text{Total AT/pole} - AT_d/\text{pole} \\ &= 6600 - 440 \\ &= 6,160 \end{aligned}$$

$$\text{ii) Additional field current} = \frac{440}{1100} = 0.4 \text{ A}$$

3. Two transformers operating in parallel have the following characteristics

No. 1 - % IR = 1.0% and % IX = 5.0%

No. 2 - % IR = 1.5% and % IX = 4.0%

How will they share a load of 100KVA at 0.8 pf lagging?

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Solution

Total load in KVA = 100

Power factor, $\cos \phi = 0.8$

phase angle of load, $\phi = \cos^{-1} 0.8 = -36.87^\circ$

Hence total load to be shared, $S = 100 \angle -36.87^\circ$ KVA

Percentage impedance of transformer A, $Z_A = (1 + j5)$
 $= 5.099 \angle 78.69^\circ$

Percentage impedance of transformer B, $Z_B = (1.5 + j4)$
 $= 4.272 \angle 69.44^\circ$

$$\begin{aligned} Z_A + Z_B &= (1 + j5) + (1.5 + j4) \\ &= (2.5 + j9) \\ &= 9.34 \angle 74.48^\circ \end{aligned}$$

Load shared by transformer A,

$$S_A = S \frac{Z_B}{Z_A + Z_B} = 100 \angle -36.87^\circ \times \frac{4.272 \angle 69.44^\circ}{9.34 \angle 74.48^\circ} \text{ KVA}$$

$$S_A = 45.74 \angle -41.91^\circ \text{ KVA or } 34.04 \text{ kW}$$

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Load shared by transformer B,

$$S_B = S \frac{Z_A}{Z_A + Z_B} = 100 \angle -36.87^\circ \times \frac{5.099 \angle 78.69^\circ}{9.34 \angle 74.48^\circ}$$

$$S_B = 54.59 \angle -32.66^\circ \text{ kVA or } 45.96 \text{ kW}$$

4. Determine the number of turns on each commutating pole of a 6-pole machine, if the flux density in the air-gap of the commutating pole = 0.5 Wb/m^2 at full load and the effective length of the air-gap is 4 mm . The full-load current is 500 A and the armature is lap-wound with 540 conductors. Assume the ampere turns required for the remainder of the magnetic circuit to be one-tenth of that the air gap.

Solution

$$N_{cp} I_a = \frac{Z I_c}{2P} + \frac{B_g l_g}{\mu_0}$$

where N_{cp} = No. of turns on the compole

I_a = Armature current

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Z = No. of armature conductors

I_c = Coil current

P = No. of poles

l_g = Air-gap length under the compole

$Z = 540$

$I_a = 500A$

$I_c = \frac{500}{6} A$

$P = 6$

$$\therefore \text{Armature mmf} = \frac{540 \times \left(\frac{500}{6}\right)}{2} \times 6 = 3750$$

$$\text{Armature mmf} = 540 \times \frac{(500/6)}{2 \times 6} = 3750$$

$$\begin{aligned} \text{Compole air gap mmf} &= B_g \times l_g / \mu_0 \\ &= 0.5 \times 4 \times 10^{-3} / 4\pi \times 10^{-7} \\ &= 1591 \end{aligned}$$

mmf required for the rest of the magnetic circuit = 10% of 1591 = 159

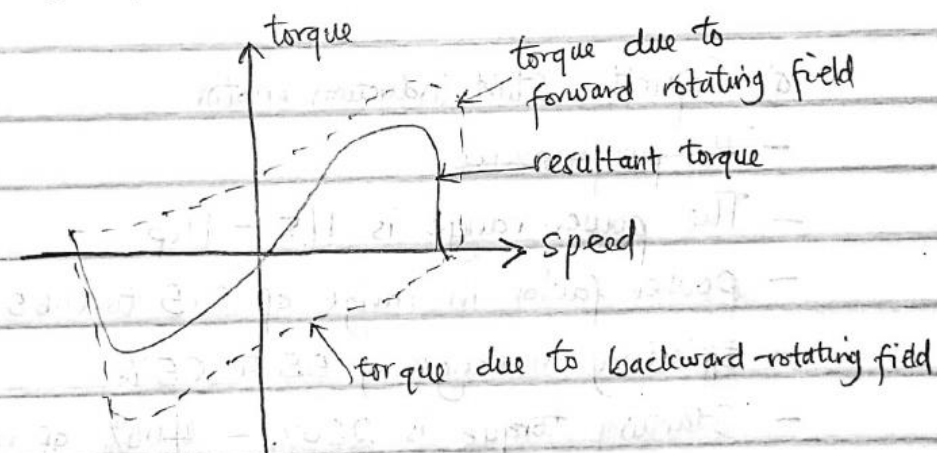
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$$\therefore \text{Total compole air-gap mmf} = 1591 + 159 = 1750$$

$$\text{Total mmf required} = 3750 + 1750 = 5500$$

$$\therefore N_{cp} I_a = 5500 \Rightarrow N_{cp} = \frac{5500}{I_a} = \frac{5500}{500} = 11$$

23) Draw and explain briefly the torque-speed characteristic of a single-phase induction machine.



When the rotor rotates, $Z_f > Z_b$ and consequently

$E_f > E_b$. Therefore, as the speed increases, the resultant forward air gap field increases while the resultant backward air gap field decreases. Hence, the forward torque increases and the backward torque decreases. ~~The torque speed characteristics~~

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24) Describe & briefly the performance evaluation of the following single-phase induction motors:

- a) Capacitor start induction motor
- b) capacitor run induction motor
- c) Capacitor start capacitor run induction motor
- d) Split-phase induction motor
- e) Shaded pole induction motor

a) Capacitor start induction motor

~~The performance~~

- The power range is $1/8 - 1$ hp
- power factor in range of 0.5 to 0.65
- Efficiency in range of 55 to 65%
- Starting torque is 250% - 450% of rated torque
- Maximum torque or break-down torque is 350% rated torque
- A typical capacitor value is $300 \mu F$ for $1/2$ hp motor
- Cost is 1.25p4

Application

- Compressors, pumps, air-conditioning, electric refrigerators and loads require heavy starting torque

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b) Capacitor run induction motor

- The power range is $1/8 - 1\text{hp}$
- Power factor in range of 0.75 to 0.9
- Efficiency in range of 60 to 70%
- Starting torque is 100% - 250% of rated torque
- Maximum torque or breakdown torque is 250% rated torque
- Capacitors are in range 20 - 50 μF
- Cost is 1.4 pu

Application

Fans, Blowers, low noise applications

Capacitor start Capacitor Run Induction Motor

- The power range is $1/8 - 1\text{hp}$
- Power factor in range of 0.75 to 0.9
- Efficiency in range of 60 to 70%
- Starting torque is 200% - 300% of rated torque
- Maximum torque or breakdown torque is 250% rated torque
- Capacitors are in range 20 - 50 μF
- Cost is 1.8 pu

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Applications

Fans, blowers, low noise and high starting torque applications

d) Split-Phase Induction motor

- The power range is $1/20 - 1$ hp
- Power factor in range of 0.5 to 0.65
- Efficiency in range of 55 to 65%
- Starting torque is 100% - 250% of rated torque
- Maximum torque or breakdown torque is 300% rated torque
- Cost is 1 pu

Applications

Fans, blowers, centrifugal pumps, washing machines, loads with low or medium starting torque

e) Shaded-Pole Induction motors

- The power range is $1/200 - 1/20$ hp
- Power factor in range of 0.25 to 0.4
- Efficiency in range of 25 to 40%
- Starting torque is 40% - 60% of rated torque
- Maximum torque or breakdown torque is 140% rated torque

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— Cost is 0.6 pu

Applications

Fans, toys, hair dryers and loads require low starting torque

25) Explain the behavior of synchronous generator when connected to (i) Isolated ~~bus~~ loads (ii) Infinite bus

The behaviour of synchronous generator when connected to:

(i) Isolated loads:

— with a lagging pf

a) current lags the terminal voltage, E

b) the voltage drop across the synchronous reactance, E_x , leads the current by 90°

c) the induced voltage, E_o , generated by the flux, Φ , is equal to the phasor sum of E and E_x

— with a leading pf

a) current leads the terminal voltage, E

b) the voltage drop across the synchronous reactance, E_x , leads the current by 90°

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c) the induced voltage, E_o , generated by the flux, Φ , is equal to the phasor sum of E and E_x

(ii) Infinite bus

Connecting a generator to other generators is called paralleling and ^{many} paralleled generators behave like an infinite bus where the voltage and frequency are constant and cannot be easily altered.

26) A 13.8-kV 10-MVA 0.8 pf-lagging 60 Hz two-pole Y-connected steam-turbine generator has a synchronous reactance of 12Ω per phase and an armature resistance of 1.5Ω per phase. This generator is operating in parallel with a large power system (infinite bus).

a) What is the magnitude of E_A at rated conditions?

b) What is the torque angle of the generator at rated conditions?

c) If the field current is constant, what is the maximum power possible out of this generator? How much reserve power or torque does this generator have at full load?

d) At the absolute maximum possible, how much reactive power will this generator be supplying or consuming? Sketch

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the corresponding phasor diagram. (Assume I_f is still unchanged)

Solution

(a) The phase voltage of this generator at rated conditions is

$$V_\phi = \frac{13800\text{ V}}{\sqrt{3}} = 7967\text{ V}$$

The armature current per phase at rated conditions is

$$I_A = \frac{S}{\sqrt{3} V_T} = \frac{10,000,000\text{ VA}}{\sqrt{3} (13800\text{ V})} = \underline{418\text{ A}}$$

Therefore, the internal generated voltage at rated conditions is

$$E_A = V_\phi + R_A I_A + j X_s I_A$$

$$E_A = 7967 \angle 0^\circ + (1.5 \Omega)(418 \angle -36.87^\circ \text{ A}) + j(12.0 \Omega)(418 \angle -36.87^\circ \text{ A})$$

$$E_A = \underline{12,040 \angle 17.6^\circ \text{ V}}$$

$\therefore$ The magnitude of E_A is 12,040 V

b) The torque angle of the generator at rated conditions is $\delta = 17.6^\circ$

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c) Ignoring R_A , the maximum output power of the generator is given by

$$P_{\max} = \frac{3V_{\phi} E_A}{X_s} = \frac{3(7967\text{V})(12,040\text{V})}{12\Omega} = 24.0\text{MW}$$

The power at maximum load is 8MW, so the maximum output power is three times the full load output power.

(d) The phasor diagram at these conditions is shown below:

Under these conditions, the armature current is

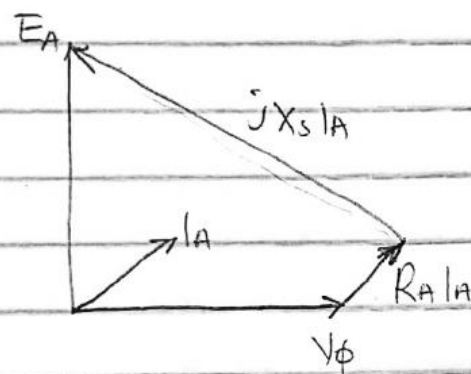
$$I_A = \frac{E_A - V_{\phi}}{R_A + jX_s} = \frac{12040\angle 90^\circ\text{V} - 7967\angle 0^\circ\text{V}}{1.5 + j12.0\Omega} = 1194\angle 40.6^\circ\text{A}$$

The reactive power produced by the generator at this point is

$$Q = 3V_{\phi} I_A \sin\theta = 3(7967\text{V})(1194\text{A}) \sin(0^\circ - 40.6^\circ) = 18.6\text{MVar}$$

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The generator is actually consuming reactive power at this time.



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